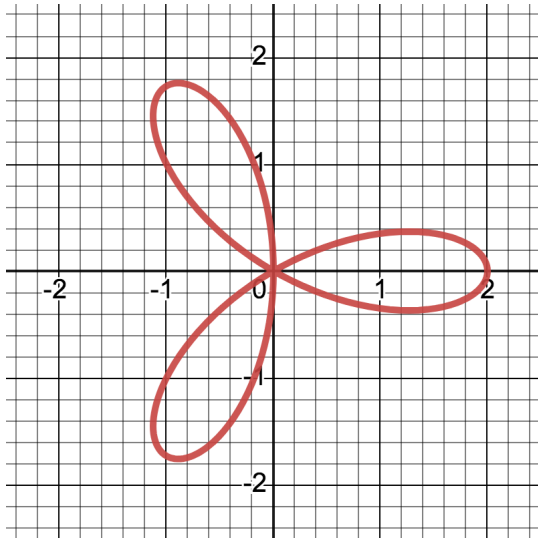


SECTION 7.4: AREA IN POLAR COORDINATES

- (1) Suppose $r = f(\theta)$ is a continuous and nonnegative on the interval from $\alpha \leq \theta \leq \beta$, then the area bounded by $r = f(\theta)$ and the radial lines $\theta = \alpha$ and $\theta = \beta$ is

- (2) Set up and evaluate the integral to find the area enclosed by the polar curve $r = 10 \cos(\theta)$.

- (3) Let R be the region enclosed by the polar curve $r = 2 \cos(3\theta)$. Shade the region R , then Set up and evaluate the integral to find the area of R .



- (4) Set up the integral to compute the area of the region inside $r = 2 \cos(3\theta)$ and outside $r = 1$.

- (5) Set up the integral to compute the area of the region inside *both* $r = 2 \cos(3\theta)$ and $r = 1$.